ON THE INTERACTION OF ESD, NBTI AND HCI IN 65NM TECHNOLOGY

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ABSTRACT

A comprehensive study on the interaction between ESD, NBTI and HCI on silicide blocked (SBLK) PFET devices is presented for a state-of-the-art 65nm bulk technology. ESD behavior of thin and thick oxide devices are shown to have opposite channel length dependence. The study of NBTI-ESD interaction on thin oxides devices shows that non-destructive ESD pre-stressing worsens the NBTI degradation. On the other hand NBTI pre-stressed thick oxide devices show high on-resistance during ESD characterization. It is shown that in thin oxide long channel length devices at high temperature pure NBTI is the worst case degradation mode whereas in short channel length devices combined "HC-NBTI" degradation dominates. Furthermore, we observed that while a SBLK PFET is HC stressed at high temperature then NBTI also takes place simultaneously, resulting in "HC-NBTI" co-activation, which is found to be channel length dependent. Finally, we have shown that HC degradation is worse at high temperature than at room temperatures due to this NBTI co-activation. [*Keywords:* ESD, NBTI, HCI, Interaction.]

INTRODUCTION

In "self-protecting" Electro Static Discharge (ESD) strategies, silicide blocked (SBLK) FETs are used both as the output driver as well as an ESD protection device, utilizing the drain/source ballast resistance to improve the parasitic bipolar junction transistor (pBJT) efficiency during an ESD event. During an ESD event, non-destructive pulses can potentially induce latent damage effects into the output driver whereas during the normal operation of the circuit it is being subjected to various kinds of electrical stress. Previous studies have concentrated on latent damage in NMOS devices due to ESD pre-stressing [1-6]. Aur et al [1] have shown that latent damage due to ESD pre-stressing could decrease the hot carrier injection (HCI) lifetime by as much as a factor of 4. It has also been reported by Doyle et al [2] that NMOSFETs show series resistance like changes in the characteristics due to the injection of primary charge carriers in the oxide during high current stress. More recently, Salman et al [6] studied the effect of non-destructive ESD stress on deep submicron devices and concluded that ESD pulses cause a certain amount of Hot Electron (HE)-like degradation. This work focuses on the interaction of ESD, NBTI and HCI in SBLK PFETs, which are becoming very important in PFET based self-protected ESD protection circuits, as the reduction in channel length causes an increase in the gain of the pBJT [7]. Due to the dual functionality of the pull up PFET, it can be subjected either to HCI or NBTI (Negative Bias Temperature Instability) or both during its normal operation, whereas it is subjected to high current stress during ESD. To predict lifetime accurately it is necessary to study the interaction and interdependence between these degradation mechanisms. Due to the aggressive scaling of oxide thickness compared to the operating voltage, the field across the oxide increases, thus making the PFET vulnerable to NBTI as well as HCI.

The study of these three reliability mechanisms divides this work into three sections. The first section presents the ESD characterization of bulk CMOS 65nm technology for three groups of PMOS devices, with different oxide thicknesses and channel length as a parameter [8]. It is shown that thin and thick oxide devices exhibit different normalized failure current (It2) behavior with respect to channel length. In particular, it is found that not only the high gain of the pBJT is an important factor in a short channel device in determining the ESD behavior, but also the power dissipation volume needs to be considered. The second section describes the interaction between ESD and NBTI. Since NBTI may occur before a (destructive or otherwise) ESD event or vice-versa during the lifetime of the device, it will be important to study the ESD-NBTI interaction and ascertain whether there are any detrimental effects of one mechanism on the other. It is observed that NBTI pre-stressing shows a change in the on-resistance, as extracted from TLP I-V, in thick oxide devices. On the other hand, non-destructive ESD pre-stressing is shown to worsen the subsequent NBTI degradation of thin oxide devices. The third section reports the results of HC stress of thin oxide devices at high temperature. It has been shown by E. Li et al [9] that hot carrier degradation at high temperature is worse than at room temperature in 0.25µm technology due to enhanced hole injection. However, in this study of hot carrier degradation in thin oxide devices at high temperature we observed a channel length dependent co-activation of NBTI during the HC stressing, as predicted in [10] for 0.35µm technology.

I. ANALYSIS OF ESD BEHAVIOR

ESD characteristics were studied using transmission line pulsing (TLP) measurements with an 100ns rectangular pulse having 7ns rise-time, which emulates the Human Body Model (HBM), for three different device groups from the 65nm bulk CMOS technology [8]: (a) Thin Oxide (SG) Devices (Vdd=1.0V, Tox=1.25nm), (b) Intermediate Oxide (EG) Devices (Vdd=1.5V, Tox=2.2nm) and (c) Thick Oxide (DG) Devices (Vdd=2.5V, Tox=5.4nm). In SG Devices the drain side silicide blocking length DOP=2µm and the source side SOP=0.4µm, whereas both EG and DG devices have DOP=4µm and SOP=0.4µm. Figure 1 shows the variation of It2 vs channel length for these devices. All the devices are multi-fingered with 240µm of total width where each finger had a width of 20µm.

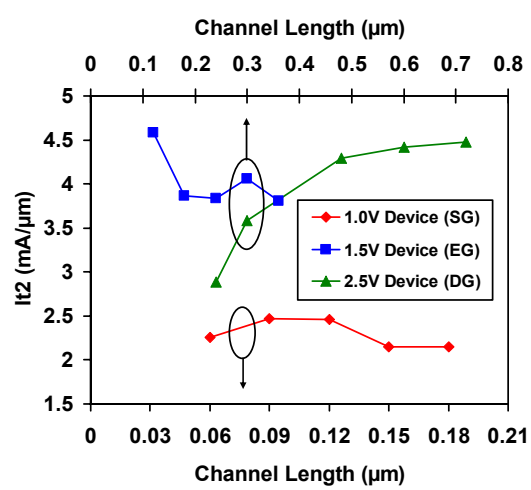

Figure 1. Dependence of It2 on Channel Length for 1.0V (Tox=1.25nm), 1.5V (Tox=2.2nm) and 2.5V (Tox=5.4nm) silicide blocked (SBLK) PMOS devices.

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It is seen from Figure 1 that the It2 of the DG devices increases with increasing channel length, while for EG devices the shortest channel length device shows the highest It2. This behavior can be explained by the fact that with respect to channel length two competing factors play a role: First, the location of the pBJT, which forms deeper into the substrate as the channel length increases [11, 12], and second, the parasitic bipolar gain (β), which increases as the channel length decreases [7]. For DG devices the pBJT forms deeper into the substrate, allowing more volume for heat to dissipate during an ESD event, resulting in higher It2 [13]. For the EG devices β dominates over the pBJT location. As the channel length decreases, β increases which in turn results in higher It2. For SG devices we expect higher It2 than for EG or DG devices, since the pBJT β increases for shorter channel length devices. However, we found that SG devices show the lowest It2. As has been argued by Bock et al [11], it may be that the power dissipation volume in the SG devices is so small, that the physical limit to hold current is reached well before the parasitic bipolar gain starts to play an effective role.

Next, we determined the breakdown mode (i.e. drain-to-source or gate oxide) of SG and DG devices under ESD stress. To determine the breakdown mode of the device, Id and Ig were compared for fresh and ESD stressed at It2 devices. As shown in Figure 2, an increase in Id and Ig shows that, in SG devices, both drain to source filamentation occurs due to the thermal heating (melting the silicon at the surface) and the oxide also gets damaged. This mode of breakdown, where both the channel as well as the oxide gets damaged, has been confirmed by measuring the various terminal currents after each TLP stress pulse, as shown in Figure 3: Id, Is and Ig all increase following ESD stress at It2.

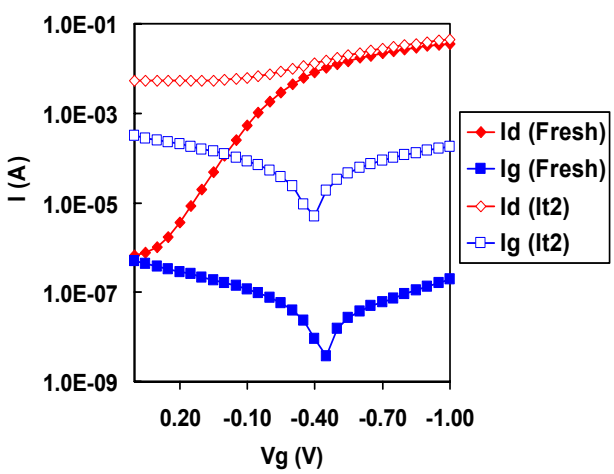

Figure 2. Id, Ig vs Vg plotted for a Fresh and an ESD stressed SG device with W/L=240/0.06. Both Id and Ig increases following stress at It2, indicating that Drain to Source filamentation occurs due to the thermal heating (melting the silicon at the surface) and that the oxide also gets damaged.

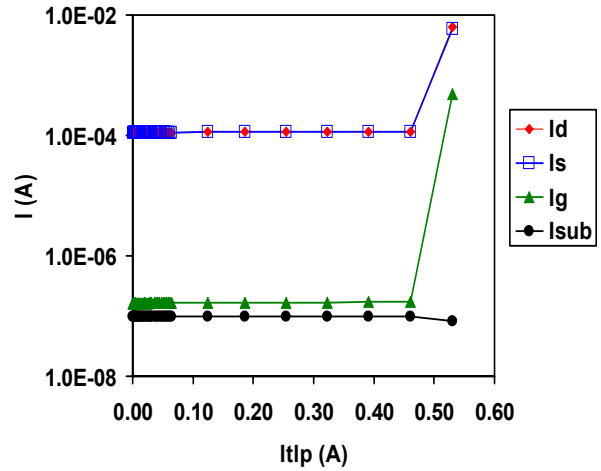

Figure 3. Terminal currents vs Itlp measured after each TLP stress pulse at Vd=-1.0V for SG device with W/L=240/0.06. Id, Is and Ig increases following stress at It2.

For DG devices, as shown in Figure 4, an increase in Id shows that only drain to source breakdown occurs. As the figure shows there is no change in Ig after It2 stress, indicating gate oxide does not get damaged due to thermal heating at breakdown. This is due to the fact that in long channel thick oxide devices the pBJT current spreads much deeper into the substrate [11] and gives more volume for heat dissipation. This is supported by Figure 5, where all the terminal currents after each TLP stress pulse are plotted vs Itlp: only Id and Is are shown to increase.

The breakdown behavior of SG and DG devices was also investigated as a function of channel length. It was found that the breakdown behavior remains the same irrespective of the channel length variation (within the present range) for both oxide thicknesses.

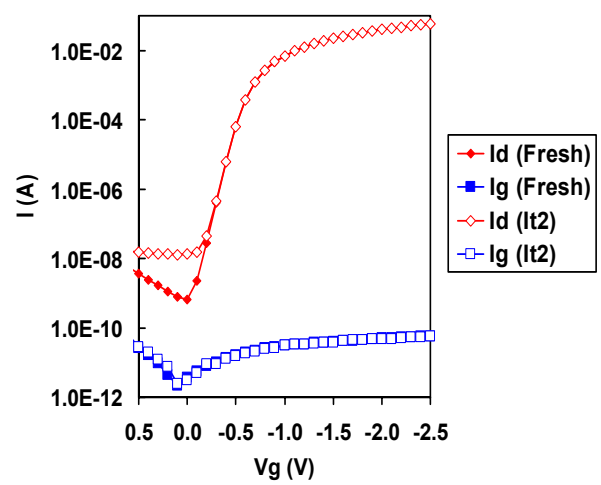

Figure 4. Id, Ig vs Vg plotted for a Fresh and an ESD stressed DG device with W/L=240/0.24. Id increases following stress at It2, indicating that only Drain to Source filamentation occurs due to the thermal heating.

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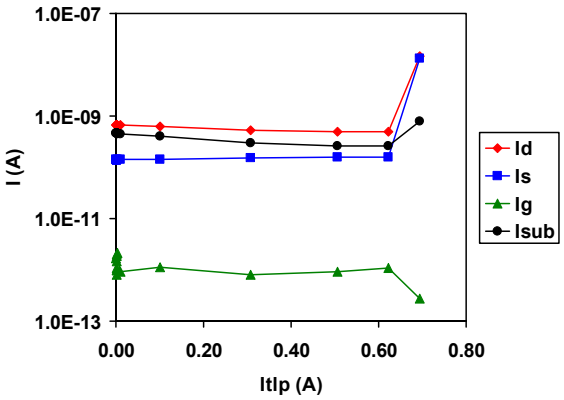

Figure 5. Terminal currents vs Itlp measured after each TLP stress pulse at Vd=-2.5V for DG device with W/L=240/0.24. Only Id and Is increases following stress at It2.

II. ESD-NBTI INTERACTION

Effect of NBTI pre-stressing on ESD characteristics

In this sub-section, we present the results on the effect of NBTI pre-stressing on ESD performance. To study this interaction the devices were first NBTI stressed at high temperature. The system was then cooled down to room temperature and subsequently TLP measurements were done. DG devices were NBTI pre-stressed at Vg=-4.5V and 140^0C for 20ksec. The chuck was then allowed to cool down to room temperature and TLP stressing was done. Figure 6 compares the TLP I-V characteristics for fresh and NBTI pre-stressed SBLK PFET with W/L=240/0.24. It can be seen that the NBTI pre-stressed device shows higher on-resistance as compared to the fresh device. The NBTI pre-stressed device also shows higher It2. To examine the reason behind the increase in resistance we have plotted the linear transconductance (GmLin) vs Vg curve for the fresh and NBTI pre-stressed device in Figure 7. The figure shows four GmLin curves from a single device: (i) Fresh device, (ii) Fresh device measured at 140^0C, (iii) NBTI pre-stressed device, measured at 140^0C, and (iv) NBTI pre-stressed device at room temperature. As shown, fresh device GmLin decreases at 140^0C and even further reduction is observed when the device is NBTI stressed. When the system is cooled down to room temperature the GmLin peak of the NBTI stressed is lower than that of fresh device i.e. NBTI pre-stressed device has a higher resistance.

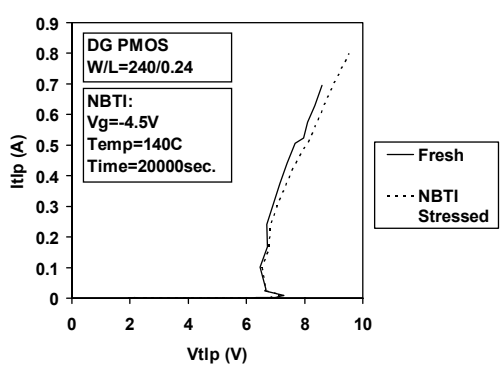

Figure 6. TLP I-V plots for a Fresh and an NBTI pre-stressed DG device. Subsequent to NBTI stressing at high temperature TLP was applied on the device at room temperature.

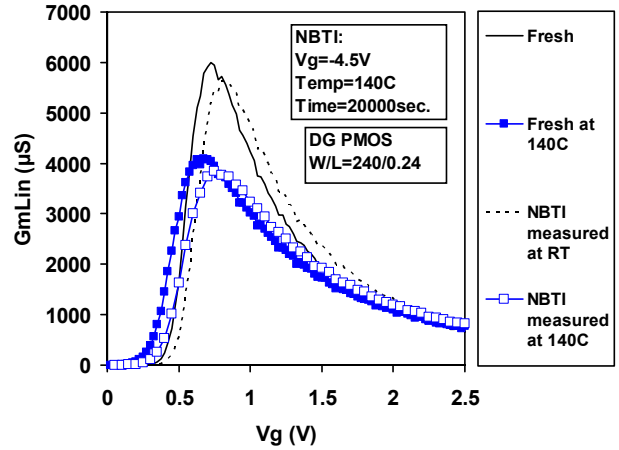

Figure 7. GmLin vs Vg plots for a Fresh and an NBTI stressed DG device with W/L=240/0.24, measured at stress temperature as well as at room temperature.

Similar studies were also done on the SG and EG devices. However, the ESD behavior of SG and EG devices showed little dependence on NBTI pre-stressing, and at this point we speculate a couple of reasons that might explain this observation. NBTI pre-stressing typically introduces oxide charge and causes interface state generation in the device under stress [14]. However, whereas the effect of both these types of damage are detected on the I-V characteristics for the thick oxide (DG) devices, for the thin oxide (SG and EG) devices only the effect of the interface states is detected. This is because carrier tunneling during the I-V measurement performance in the case of thin oxide devices neutralizes the oxide charge [15].

Secondly, the resistance of the device increases as the channel length increases. In short channel devices, the resistance of the channel region will constitute a small fraction of the total compared to what is contributed by the SBLK regions. The change in channel resistance caused by NBTI pre-stressing would therefore be small, and thus the ESD behavior due to NBTI pre-stressing does not get affected much in short channel length devices.

Effect of ESD pre-stressing on NBTI characteristics

In this sub-section, we present the impact of ESD pre-stressing on subsequent NBTI degradation behavior. This interaction has been studied only for SG devices as these devices are more prone to NBTI degradation compared to the thick oxide devices [14]. Following the non-destructive ESD pulses at drain the devices were subjected to NBTI stress. In particular, a SG SBLK PFET with W/L=240/0.06 was ESD stressed at room temperature with non-destructive pulses up to 50% of It2 (~1.13mA/µm) and following this it was subjected to NBTI stress at Vg=-1.8V at T=140^0C. It can be seen from Figure 8 that the Vtsat of the ESD pre-stressed device degrades ~10% more than when only NBTI stress is applied. This must be because high current stressing on the drain decreases the reliability of oxide by weakening the interface Si-H bonds [2], which in turn leads to higher degradation in Vtsat by subsequent NBTI stress.

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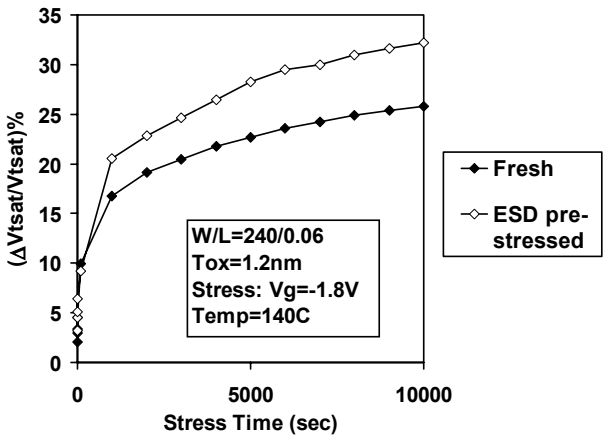

Figure 8. NBTI Vtsat(%) shift vs Stress Time, for a Fresh and an ESD pre-stressed SG SBLK PMOS device with W/L=240/0.06.

The NBTI behavior was also studied for SG SBLK PMOS devices with channel length as a parameter. It can be seen from Figure 9 that the nominal channel length device (L=0.06µm) degrades more as compared to longer channel length devices, across which there is little variation in the Vtsat degradation. Ideally, NBTI degradation should be independent of channel length, as it is determined by gate voltage alone. However, it has been pointed out by Yamamoto et al [16] that for short channel length devices, localized degradation of the oxide near the gate edge, due to electrochemical reactions between holes and initial inactive oxide defects, can increase the NBTI for short channel length devices. Also, Boron penetration from the Source and Drain diffusions into the oxide above the channel causes generation of positive charges by electrical stress over an area which for short channel devices is a considerable part of the channel [14, 16].

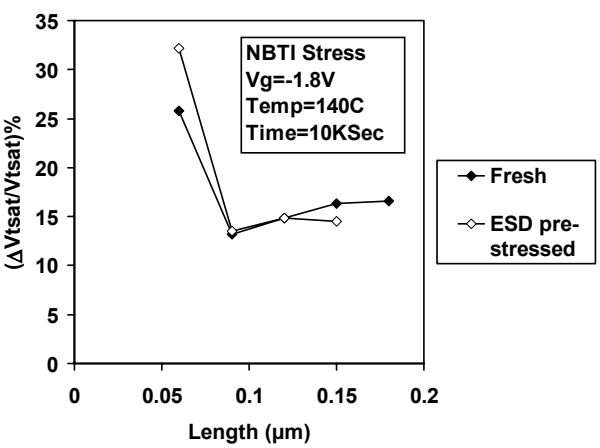

Figure 9. NBTI Vtsat(%) shift for Fresh and ESD pre-stressed SG SBLK PMOS devices vs Channel Length.

Figure 9 also shows the effect of ESD pre-stressing (~50% of It2) on NBTI behavior of the various channel lengths devices from the SG group. It is found that NBTI performance due to ESD pre-stressing gets affected only for the nominal channel length device. For longer channel length devices the ESD pre-stressing does not have significant effect on NBTI characteristics. This is because while for the nominal channel length device the turn-on region of pBJT is along the gate oxide/channel interface, for the longer channel length devices it moves to the bottom of diffusion region [11, 12]. This prevents the oxide interface from being weakened by the ESD pre-stress for the long channel devices.

III. HCI-NBTI INTERACTION

NBTI and HC stressing were performed to identify the worst case degradation mechanism, either HCI or NBTI, for SBLK PFETs. We chose 240µm wide SG SBLK PFET with channel length as a parameter. The three channel lengths of these devices are 0.06µm, 0.09µm, and 0.12µm. Apart from the channel length difference among these devices all the other processing conditions were kept identical. The devices were stressed at 140^0C under two different bias conditions: (a) Vg=Vd=-1.8V and (b) Vg@I_{sub}(max.), Vd=-1.8V. Figures 10, 11 and 12 show the saturation threshold voltage change (ΔVtSat) vs Stress Time for L = 0.06µm, 0.09µm and 0.12µm respectively. As can be seen from these figures, worst case degradation occurs at stress condition Vg=Vd and not at the maximum substrate current. An interesting observation from these figure is that the degradation at 140^0C in the case of L=0.09 and 0.12µm devices (Figs. 11 and 12) for stress condition Vg=Vd parallels the pure NBTI degradation. The similar time dependence between NBTI and HCI (Vg=Vd stress) shows that at high temperature under Vg=Vd stress condition concurrent "HC-NBTI" activation occurs. It is also observed from Figures 11 and 12 that pure NBTI dominates over HC stress condition in these devices. However, for the short channel device (L=0.06µm), as shown in Figure 10, even though initially NBTI dominates over HC, the degradation rate of "HC-NBTI" is higher than pure NBTI. Thus channel length dependent degradation mode at high temperature under Vg=Vd stress condition reveals that an "NBTI-like" phenomenon dominates in long channel devices whereas "HC-NBTI" dominates in short channel devices.

The dominance of NBTI under Vg=Vd stress condition at high temperature for the long channel devices is due to the fact that a large portion of channel near the source side remains populated with holes. These holes which cover a substantial portion of the channel cause NBTI like degradation near the source whereas near the drain the oxide is damaged by hot carriers [10]. By contrast, in short channel devices the degradation is the result of the "combined" HCI and NBTI. The results for L = 0.09µm and 0.12µm are in agreement with the observation by E. Li et al [9] who argue that enhanced hole injection into the gate oxide at Vg=Vd is the cause of worst case degradation.

Guerin et al [17] have found that even at room temperature the degradation time dependence under Vg=Vd stress condition is same as that for NBTI stress. Hence, the HCI stresses were also done at room temperature and are plotted in Figures 10, 11 and 12 for L = 0.06µm, 0.09µm and 0.12µm respectively. For our SBLK PMOS devices, we found that degradation at room temperature is negligible for all the channel length devices, and stress voltages examined.

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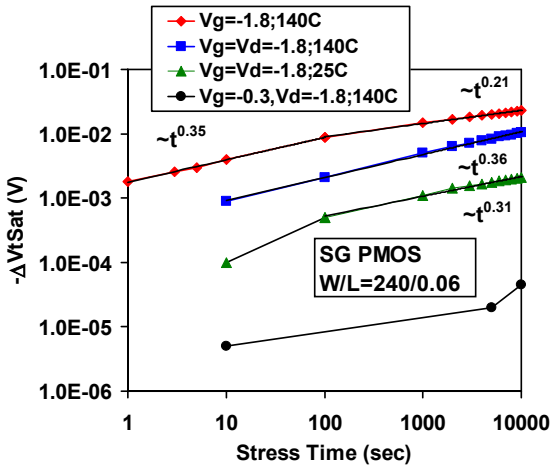

Figure 10. Saturation Threshold Voltage shift ΔVtSat vs Stress Time for different stresses for SG devices with W/L=240/0.06.

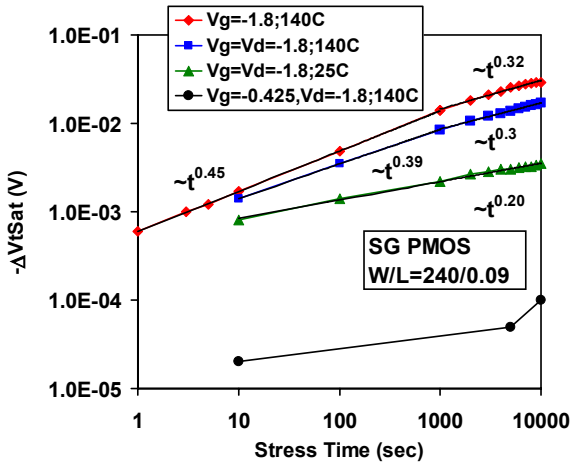

Figure 11. Saturation Threshold Voltage shift ΔVtSat vs Stress Time for different stresses for SG devices with W/L=240/0.09.

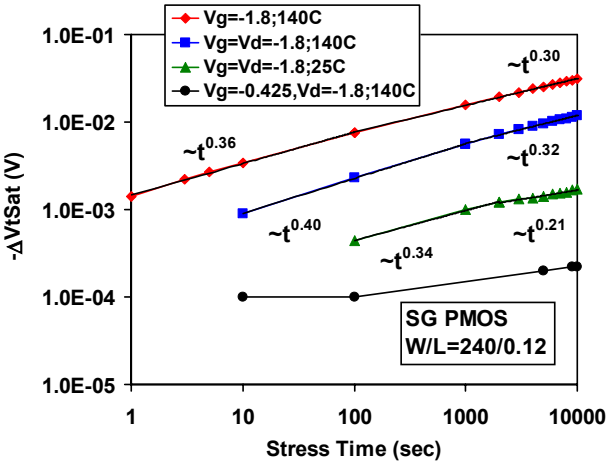

Figure 12. Saturation Threshold Voltage shift ΔVtSat vs Stress Time for different stresses for SG devices with W/L=240/0.12.

CONCLUSION

A comprehensive study on the interaction between ESD, NBTI and HCI on SBLK PFET devices is presented for a state-of-the-art 65nm bulk technology, with three oxide thickness values and channel length as a parameter. It was found that thin and thick oxide devices exhibit different It2 behavior with respect to channel length. It is pointed out that, as channel length decreases small heat dissipation volume becomes a major performance limiting factor, even though the pBJT gain increases as channel length shortens. The interaction between ESD and NBTI showed that NBTI pre-stressed thick oxide devices exhibit high on-resistance during subsequent ESD characterization. On the other hand, when the device is pre-stressed with non-destructive ESD pulses the NBTI characteristics of the thin oxide device degrade. The Vtsat shift increases by ~10% due to ESD pre-stressing as compared to NBTI only stressed device. NBTI studies on thin oxide devices revealed channel length dependent behavior, whereby the shortest channel length device shows the highest degradation. Finally, it has been shown that pure NBTI is the worst case degradation condition for thin oxide long channel devices, whereas in short channel length devices (L=0.06µm) "HC-NBTI" degradation dominates. Also, HC stressing at high temperature reveals that for longer channel length devices there is simultaneous NBTI degradation.

ACKNOWLEDGEMENT

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